U-P-001250 (DSA-DM)
1 November 1966

INTERIM FEDERAL SPECIFICATION

PULP PROTECTOR, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a hard setting, dental pulp protector suitable for use as a rigid base under restorative materials.

1.2 Classification. Dental pulp protector shall be of the following types, as specified (see 6.1):

Type I. - Dental pulp protector (with mixing and applying instrument).

Type II. - Dental pulp protector replacement (without instrument).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

FSC 6520

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Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The dental pulp protector shall consist of a base paste which, when mixed with catalyst paste in accordance with instructions (see 3.10) accompanying the package, produces a non-toxic, self-setting, pulp protector suitable for use as a rigid base under silicate cement, amalgam and resin filling materials.

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3.1.1 Base paste. The base paste shall consist of the reactive agent 1,3 butylene glycol disalicylate combined with inert pigments of titanium dioxide, calcium sulfate and ferric oxide. Calcium tungstate shall be added as a radiopaquing agent.

3.1.2 Catalyst paste. The catalyst paste shall consist of a reactive ingredient, calcium hydroxide; a filler, zinc oxide, and a dispersing or suspension agent, zinc stearate, suspended in a suitable organic vehicle. The calcium hydroxide shall be in stoichiometric excess over amount required to react with 1,3 butylene glycol disalicylate.

3.2 Description. The pulp protector base paste and catalyst paste shall be supplied in airtight, collapsible, metal composition tubes. In addition, the tubes shall be of sufficient strength and flexibility that leaking shall not occur during normal use. The pulp protector shall be radiopaque to permit differentiation from dentin or caries.

3.2.1 Accessories. The type I unit package shall be furnished with a mixing pad and a mixing and placement instrument. The type II unit refill package shall be furnished with a mixing pad without the mixing and placement instrument.

3.2.1.1 Mixing pad. The mixing pad shall be compatible for use in mixing pulp protector materials without absorption or contamination, and be of sufficient thickness to prevent tearing. The pad shall contain a minimum of 90 sheets. Each sheet shall be 1-3/8 inches wide by 2-3/4 inches $\pm$ 1/16 inch long and 0.003 inch $\pm$ 0.001 inch in thickness. The pad shall be glued on two adjacent sides to insure firm support during manipulation of mix, yet shall permit easy removal when used as intended. In addition, each pad shall be mounted on a suitable cardboard base.

3.2.1.2 Mixing and placement instrument. The mixing and placement instrument shall be fabricated of austenitic corrosion-resistant steel. The instrument shall have the following dimensions with a $\pm$ 5.0 percent tolerance: the overall length, 4-1/4 inches; the handle length, 3-1/2 inches; handle diameter, 1/8 inch and the tip length, 1/4 inch bent at an angle of 35 $\pm$ 5 degrees. The tip shall taper to the distal end and shall terminate in a small ball 0.030 $\pm$ 0.003 inches in diameter. The handle shall be satin-finished to provide a finger grip, without knurling. Other portions of the metal shall be highly polished to facilitate cleaning.

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3.3 Physical properties.

3.3.1 Working time. The working time of the pulp protector shall be a minimum of 2.0 minutes, when tested as specified in 4.4.2.

3.3.2 Setting time. The setting time of the pulp protector shall be 2.5 ± 0.5 minutes when tested as specified in 4.4.3. The mixture shall set rapidly in the mouth when used as intended.

3.3.3 Consistency. The consistency of the base and catalyst pastes shall be 33.0 ± 3.0 millimeters (mm.) and 38.0 ± 3.0 mm. respectively, when tested as specified in 4.4.4.

3.3.4 Toxicity. The manufacturer shall furnish prior to delivery, a statement directly to the Directorate of Medical Material (Code ATSB-3), Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, that the pulp protector, when clinically tested and used according to the directions supplied with the material, has no toxic effect on a normal, healthy person. (see 6.1 (d)).

3.3.5 Uniformity. The pulp protector shall have a smooth creamy appearance and shall be free of lumps and granules.

3.3.6 Leaking. The immediate containers shall pass the leak tests specified in 4.4.5.

3.4 Amount of material in unit containers. The pulp protector shall be furnished in two tubes. The base paste tube shall contain a minimum of 12.5 grams and the catalyst tube shall contain a minimum of 10.5 grams.

3.5 Compatibility. The immediate containers and all components shall be compatible, and shall not interact physically or chemically with the base paste or catalyst paste.

3.6 Cleanliness. The immediate container shall be free from all dirt, foreign matter, or other contaminants, both before and after filling.

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3.7 Stability. The pulp protector shall comply with this specification for at least three years from the date of delivery to the Government, when stored at temperatures of 70° to 90° Fahrenheit (F.).

3.8 Delivery schedule. Not more than 3 months shall have elapsed from the date of manufacture of the pulp protector to the date of delivery to the Government.

3.9 Marking. Each tube and unit package of material shall be marked with a serial number or combination of letters and numbers which shall refer to the manufacturer's records for that particular lot or batch. The year and month of manufacture shall be suitably coded as part of the lot number. In addition, each tube of pulp protector shall be permanently and legibly identified as "Base" or "Catalyst" and marked with the manufacturer's name or registered trademark.

3.10 Instructions. Adequate and accurate instructions shall accompany each package including storage instructions, method of application, and the effects of increased temperature and humidity on working and setting times. In addition, 2 copies of the instructions shall be forwarded to the contracting officer for information purposes.

3.11 Workmanship. Workmanship shall be first class throughout. The pulp protector shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the raw material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted as indicated in table I. The unit of product for inspection purposes shall be one package of pulp protector.

Table I. Sampling Criteria for Tests

Characteristics	Requirements	Test Procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations
Leak Test	3.2 & 3.3.6	4.4.5	3	1	3
Working time	3.3.1	4.4.2	1	3	3
Setting time	3.3.2	4.4.3	1	3	3
Consistency	3.3.3	4.4.4	1	6	6
Material weight	3.4	SMI.	3	2	6

4.2.3 Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The pulp protector shall be examined to determine compliance with specification requirements.

4.4 Tests.

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4.4.1 Standard testing conditions. All physical tests shall be made under uniform atmospheric conditions 70.0 ± 1.0 degrees Fahrenheit and 55.0 ± 1.0 percent relative humidity, unless otherwise specified. Equipment and material shall be conditioned in testing room for not less than 10 hours prior to making the tests.

4.4.2 Working time. A mix of pulp protector shall be prepared by combining 1.17 grams of base paste with 1.0 grams of catalyst paste. The mix shall be completed within 15 seconds on parchment pad and each 15 seconds after, the mix shall be picked up, puddled and the instrument shall be pulled away. The material shall flow together and shall not be tufted. The working time shall be the time from beginning of mix until the material does not pull together smoothly.

4.4.3 Setting time. A mix of pulp protector shall be prepared by combining 1.17 grams of base paste with 1.0 grams of catalyst paste. The mix shall be completed within 15 seconds on parchment pad and a portion shall be placed within a brass ring (9 millimeters inside diameter, 13 millimeters outside diameter and 1 millimeter deep) set on a sheet of cellulose acetate, and leveled off. This shall be done within 0.5 minutes from start of mix. The mix shall be probed with the mixing and placement instrument, and as soon as any hardening of material occurs, the pulp protector shall be considered set. Time shall be measured from start of mix.

4.4.4 Consistency. Load 0.5 cubic centimeter (cc.) of base paste or catalyst paste into a glass tube measuring device by extrusion. The material shall be placed on a sheet of cellulose acetate and covered with a second sheet. A 20 gram glass plate and a 50 gram weight shall be placed on the second sheet for 1.0 minute. The major and minor diameters of the base paste and catalyst paste shall be measured. The average of three tests shall be as specified in 3.3.3 for each material.

4.4.5 Leak test. Leak tests on the immediate containers with contained material shall be conducted in accordance with PPP-C-186.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. For type I pulp protector, one (1) tube of paste, one (1) tube of catalyst, one (1) mixing pad and one (1) instrument, as specified, constitutes one unit of issue. For type II one (1) tube of paste, one (1) tube of catalyst, and one (1) mixing pad, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate container. Immediate container.

5.1.3.1.1 Type I. Immediate container for type I pulp protector, shall be a clear transparent plastic box with a hinge lid and suitable closure. Inside dimensions of the box shall be 4-5/8 inches in length, 3-9/16 inches in width, and 1 inch in height. Each box shall be fitted with a tray insert fabricated from polystyrene foam. The tray shall have die cut outs to accommodate two tubes, one mixing pad, and one instrument. Tray dimensions shall be 4-1/2 inches in length, 3-1/2 inches in width, and 15/16 inch in height.

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5.1.3.1.2 Type II. Immediate container is not required. The type II pulp protector shall be provided with the tray insert as described under type I.

5.1.3.2 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.3 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard box.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

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5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Type required (see 1.2).

(c) Selection of applicable levels of packing. (see 5.1 and 5.2).

(d) For non-military activities, where statement to be sent (in lieu of Defense Personnel Support Center).

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6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

U-P-001250 (DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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